

Pulkit Dhingra

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📍 Chennai, India

Innovative engineer with hands-on expertise in Python and SQL, complemented by deep experience in cloud platforms (AWS, GCP, Azure) and infrastructure automation. Skilled in architecting intelligent systems, building end-to-end ML pipelines, and leveraging CI/CD, containerization, and automation to drive scalable solutions. Proven ability to translate complex technical challenges into elegant solutions, collaborate effectively with cross-functional teams, and mentor peers. Passionate about emerging technologies in AI and ML, continuous learning, and fostering collaborative, high-impact work environments.

EDUCATION

University of Bristol, Bristol, United Kingdom Nov 2025
MSc Data Science

- Analysed energetic particle measurements from live meteorological balloon flights, performing instrument calibration and correction, generating altitude and latitude-dependent flux profiles, and validating outputs against satellite observation data -- hands-on work with real physical sensor systems and measurement pipelines.
- Applied Machine Learning and Deep Learning models to spectroscopic sensor data from crop plants, building end-to-end pipelines from raw instrument output to validated nutrient estimation results.

Dr A.P.J Abdul Kalam Technical University, Lucknow, India Jun 2022
Bachelor of Technology Computer Science

- Maintained a GPA of 7.9/10, demonstrating a strong understanding of the material and a dedication to academic excellence.
- Built and delivered an end-to-end software solution under competition conditions, including data ingestion pipelines, RESTful APIs, and fine-tuned ML models -- placing first across the cohort.

WORK EXPERIENCE

Ford Motor Company (2+ years)

AI Engineer, Chennai Jan 2026 - Present

- Designed and built production-grade agentic automation systems with full architecture covering state management, memory handling, and multi-component orchestration – owning component design from requirements through to deployment.
- Developed software pipelines integrating multiple vehicle data sources to surface diagnostics, warranty, and recall insights; designed and deployed ML models for demand forecasting with evaluation frameworks to validate system behaviour and define acceptance criteria.
- Authored comprehensive test suites covering unit, integration, stress, and AI-driven evaluation cases; executed end-to-end validation using Selenium, SonarQube, and DeepEval to verify system behaviour across functional and edge case scenarios.
- Implemented LLM-as-a-judge frameworks for automated trace evaluation and integrated Arize Phoenix into production pipelines for real-time log analysis and anomaly detection; designed AI agent evaluation pipelines with Guardrails to define safety, reliability, and compliance test criteria.
- Developed performance benchmarking and system modelling frameworks to analyse AI pipeline behaviour under varying load conditions, identifying bottlenecks and tracking model drift to ensure reliability across operational scenarios.

Data Scientist, Chennai Jan 2023 - Aug 2024

- Analysed connected vehicle and supply chain data to develop algorithms identifying operational trends; processed and analysed high-volume telemetry data from embedded automotive systems, applying statistical models and signal analysis techniques to extract patterns supporting inventory, pricing, and operational decisions.
- Built automated validation and auditing workflows using LLM and ML models to review contract and vehicle documentation against backend data sources, flagging discrepancies and surfacing anomalies – generating \$100M in identified overpayment value.
- Planned and executed task-specific unit and evaluation test cases for LLM-based systems, validating response correctness through automated validation pipelines; participated in full Agile sprint cycles in Jira, collaborating with stakeholders to define acceptance criteria and evaluate feature delivery.
- Designed and maintained CI/CD pipelines using GitHub Actions, Cloud Build, and Kubernetes, integrating automated test execution and Terraform-managed infrastructure provisioning to support reliable delivery of data and ML services.

Nagarro, DevOps Engineer, Lucknow

(9 months) May 2022 – Jan 2023

- Designed and implemented CI/CD pipelines using Jenkins and Git across Python, Maven, and npm-based applications; integrated multi-cloud deployment workflows across Azure, AWS, and GCP, automating infrastructure provisioning and reducing deployment errors by validating pipeline integrity at each integration stage.
- Built and validated Infrastructure as Code (IaC) using Terraform scripts for automated cloud infrastructure provisioning, ensuring consistent and reproducible environment configuration across development and staging environments while enhancing operational efficiency.
- Implemented containerisation solutions using Docker and Kubernetes, standardising deployment environments and achieving a 35% improvement in scalability and resource utilisation while reducing configuration drift across regression testing.
- Analysed build and deployment logs on Linux to trace root causes of failures, validating pipeline integrity and improving code quality by 40% through systematic troubleshooting and end-to-end validation across integration stages.

SKILLS

Technical Skills: Applied Machine Learning, AI Agentic Automation, System Design, Algorithm Development, Signal Processing, Data Analysis, MLOps, Model Monitoring, Model Evaluation, Prompt Engineering, RAG Systems, Cloud Computing, CI/CD Pipelines, Agile/Scrum.

Programming Skills: Python, SQL, Shell Scripting, TypeScript.

Frameworks and Tools: TensorFlow, PyTorch, scikit-learn, XGBoost, LangChain, Google ADK, LangGraph, Anthropic SDK, FastAPI, DeepEval, Guardrails, Pytest, MLflow, Streamlit, FAISS, Chroma, Git, Docker, Kubernetes, GitHub Actions, Terraform, SonarQube, Selenium, Jira, AWS, Google Cloud Platform, Azure, BigQuery.

CONTRIBUTIONS AND RESEARCH

[@Byte-Sized-Brilliance-AI](#) (Start-up Initiative) [@GeeksForGeeks](#)

Feb 2025 - Present

- Founder and author at Byte Sized Brilliance AI with 30+ articles covering ML algorithms, agentic AI systems, and applied Python. Technical Writer at GeeksForGeeks with 15+ articles on Python and ML.

Automated-Document-Parser – (PYPI – Open Source module)

- A document parsing module that helps in the ingestion of Documents and information from other data objects to help build the RAG pipeline. The module is active on PYPI and can be installed using pip [\[Link\]](#)

Research Papers [\[Sketch to Face\]](#) [\[Glass Identification\]](#)

Nov 2022

- Published two applied machine learning research papers: GAN-based image generation and XGBoost-driven chemical classification, presented with structured experimental methodology and data-driven analysis (2021-2022).

Certifications

- IBM Advanced Machine Learning and Signal Processing (2020)
- Microsoft Certified Azure Fundamentals (2022)

PROJECTS

Query Generation Agent [\[Link\]](#)

Agentic-AI, LangGraph, LLMs

- Built an automated SQL query generation pipeline using Qwen-Coder 7B with an AI agent layer for request interpretation, query generation, and validation against expected results and edge cases.

Scooter IMU Analytics [\[Link\]](#)

Signal processing, Pytest, Fast API

- Developed a multi-algorithm analytics platform on 3-axis gyroscope IMU data, implementing 8 signal processing algorithms (FFT-based bearing wear detection, RMS road quality, dead-reckoning), ML models for manoeuvre classification and anomaly detection, validated through a structured Pytest suite.

Hindi GPT Neo 2.7B [\[Link\]](#)

LoRA Fine Tuning LLM, Conversational AI

- Integrated lightweight LoRA adapters into GPT-Neo 2.7B fine-tuning pipeline on Wikipedia-sourced Hindi language data, creating an efficient LLM for context-aware Hindi conversation.